

A Rapid HPLC Method for the Quantification of Tyrothricin, Menthol, and Benzocaine in Pharmaceutical Formulations

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Abstract □ A rapid, sensitive, and accurate reversed-phase HPLC method has been developed for the analysis and quantification of pharmaceutical formulations containing tyrothricin (1), an antibiotic used in antiseptic buccal compressed tablets for local application. The assay has been carried out under isocratic conditions, using a stationary phase of alumina particles coated with polybutadiene and an alkaline mobile phase (pH = 8.2). No HPLC method was reported for the analysis of 1. So, this new technique is an alternative to the slow and tedious microbiological methods. On the other hand, it allows the simultaneous quantification of 1, benzocaine (2), and menthol (3), an aromatic compound not currently analyzed by liquid chromatography.

Introduction

Tyrothricin (1) is a polypeptidic antibiotic used in the formulation of buccal antiseptics for local application,¹ with a view to the elaboration of oral tablets that are meant to be sucked. It is commonly used in conjunction with 2, a topical anesthetic, and 3. Pharmacopoeias of relevant scope² propose a microbiological turbidimetric assay for the analysis of 1. An agar diffusion test has also been reported for the analysis of this substance.³ Nevertheless, these techniques are much too slow and complicated for routine analysis.

In order to decrease the time of analysis, we have developed a rapid and selective HPLC method to quantify simultaneously all the drugs present in the formulation: 1, 2, and 3. According to the bibliographical revision performed, we have not found any method for the quantification of 3 by HPLC.⁴⁻⁷ Our technique allows the detection and quantification of this substance. Compound 2 can be analyzed by HPLC or gas chromatography⁸⁻¹⁰ as well as spectrophotometry^{8,11} and colorimetry.¹² The assay has been carried out under isocratic conditions, using a stationary phase of alumina particles coated with polybutadiene and an alkaline mobile phase (pH = 8.2).

Experimental Section

Reagents—Mannitol and sorbitol were obtained from Acofarma (Tarrasa, Spain). Compounds 1-3 and gramicidin were obtained from Merck (Darmstadt, Germany). All the reagents were of guaranteed grade (Merck) and were used as received. Methanol used was HPLC grade (Panreac, Barcelona, Spain) and the water was freshly distilled.

High-Performance Liquid Chromatography—The HPLC system consisted of a constant-flow pump (Kontron Instruments, type 420), a Rheodyne type 7125 injector equipped with a 20- μ L loop, a variable-wavelength detector (Kontron Instruments, type 432), and an integrator (Konik Instruments, type DataJet 4600). The column used (E. Merck, Aluspher RP select B, 5 μ m particle size, 12.5 cm X 4 mm ID) was packed with alumina particles bonded with polybutadiene.

A flow rate of 1 mL/min for the mobile phase (methanol-water-diammonium phosphate 30:70:0.2, v/v/w, pH = 8.2) was employed and the variable-wavelength detector was set at 191 nm.

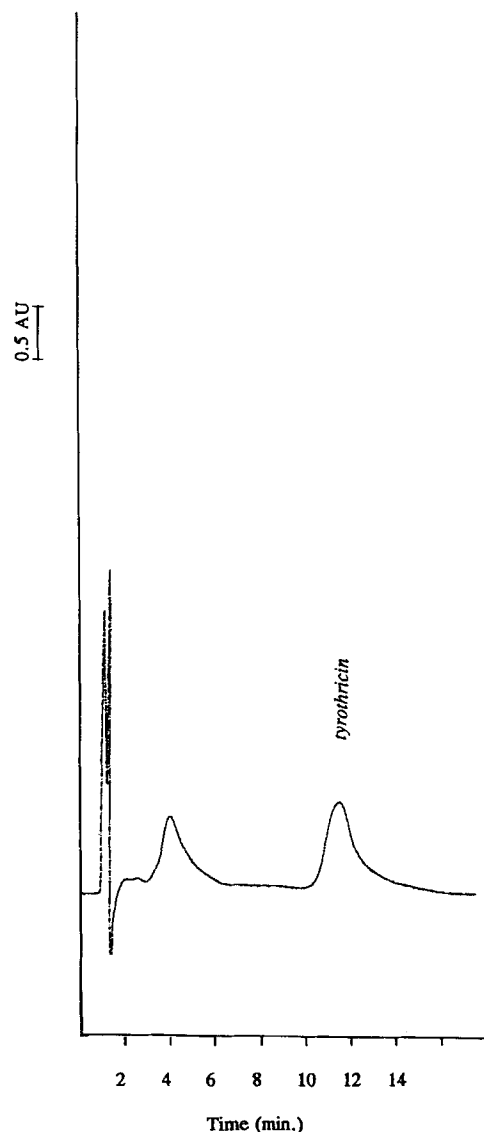


Figure 1—Chromatogram of standard tyrothricin in absolute ethanol.

Each peak area was computed automatically by the integrator. The elution was carried out under isocratic conditions at room temperature ($22 \pm 2^\circ\text{C}$).

Samples Preparation—A 2-g sample of accurately weighed powdered substances was added to 10 mL of absolute ethanol in sealed containers. After shaking (1 h) at room temperature ($22 \pm 2^\circ\text{C}$), 1 mL of the liquid was directly injected into a chromatograph by using a syringe-coupled nylon filter (Teknokroma, Barcelona, Spain).

Validation Study—The method validation was carried out by analyzing solutions containing several concentrations of 1-3 in five replicates. Furthermore, these solutions were analyzed by triplicate on five different days (15 times). The intra-assay precision was determined from the CVs of the obtained values for the samples analyzed on the

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Table 1—Intra- and Interassay Precision^a and Accuracy^b

Compound	Theoretical Concn (mg/mL)	Intra-assay (n = 5)			Interassay (n = 5)		
		Mean Area	Precision (%)	Accuracy (%)	Mean Area	Precision (%)	Accuracy (%)
Tyrothricin	25.0	0.4280	1.6	+3.3	0.4268	2.2	+3.0
	10.0	0.1566	2.7	-4.1	0.1531	4.3	-6.2
	5.0	0.0747	4.4	-6.2	0.0762	3.7	-4.5
	1.0	0.0142	4.2	+6.8	0.0144	5.3	+7.9
Benzocaine	3.0	78.376	0.8	-7.6	77.620	1.8	-8.5
	1.0	29.124	1.0	+1.3	29.264	3.4	+1.8
	0.1	3.6076	1.2	+1.7	3.6300	2.1	+2.5
	0.01	1.0576	4.9	+6.3	1.0548	6.0	+5.3
Menthol	30.0	0.3536	0.9	+1.6	0.3553	2.1	+2.1
	15.0	0.1929	3.1	+4.9	0.1978	3.8	+7.9
	7.5	0.1049	4.8	+2.2	0.1056	5.7	+3.1
	1.0	0.0333	5.8	+10.4	0.0333	6.3	+9.8

^a Precision = CV = (SD × 100)/mean. ^b Accuracy = percent deviation = (measured - theoretical) × 100/theoretical.

Table 2—Extraction Recovery (%) for Several Samples (Mean ± SD, n = 5)

Concn (mg/mL)	Extraction Recovery (%)		
	Tyrothricin	Benzocaine	Menthol
0.1	<i>a</i>	95.1 ± 2.3	<i>a</i>
0.5	<i>a</i>	96.2 ± 3.8	<i>a</i>
1.0	90.1 ± 3.9	98.1 ± 4.3	88.2 ± 6.5
3.0	92.3 ± 4.7	97.0 ± 2.8	90.1 ± 4.3
10.0	93.4 ± 2.7	<i>a</i>	92.6 ± 2.5
25.0	94.6 ± 3.0	<i>a</i>	92.4 ± 2.7
overall	92.6 ± 3.9	96.6 ± 3.5	90.8 ± 4.1

^a Not assayed.

same day. Interassay data were calculated using the mean value of the three injections performed on each day.

To determine the linearity of the detector response, calibration standard solutions of 1-3 were prepared in absolute ethanol.

The absolute extraction recoveries were evaluated at concentrations of 1, 3, 10, and 25 mg/mL for 1 and 3, and 0.1, 0.5, 1, and 3 mg/mL for 2. This evaluation was carried out by comparing the areas under the peaks of several samples with those of unextracted reference standards containing the corresponding concentrations.

Results and Discussion

The selection of the solvent used to prepare samples is critical in this analytical method. Initially, the mobile phase was used as solvent. However, absolute ethanol was finally selected for the two following reasons: 1 has a poor solubility in water; so the mobile phase, an aqueous-alcoholic solution,¹³ is saturated earlier than absolute ethanol. On the other hand, excipients commonly used in these formulations, such as mannitol¹⁴ and sorbitol,¹⁵ exhibit very low solubility in absolute ethanol. For this reason, the interference of these substances was avoided.

By injecting gramicidin and standard solutions, it has been found that, under our chromatographical conditions, gramicidin elutes near the solvent front (1.07 min). Due to this circumstance and to the fact that gramicidin is a minor constituent of 1, the peak corresponding to tyrocidine has been chosen for the quantification of 1.

The use of a very low wavelength (191 nm) has been necessary due to the fact that 1 is only detectable at this wavelength. Therefore, special precautions were taken in order to prevent the loss of precision and accuracy: first, all the employed solvents and reagents were of analytical grade in order to reduce the potential interfering substances. On the other hand, special care was taken in order to prevent a possible uneven evaporation

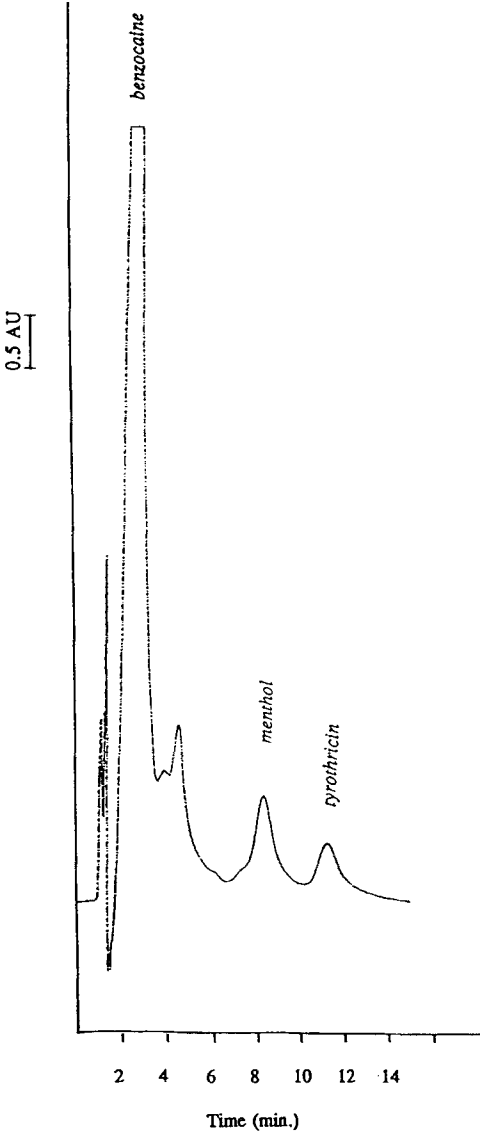


Figure 2—Chromatogram of one of our formulations.

of samples, standards, and mobile phase: storage time for these solutions were as low as possible, these phases being prepared and deaerated under homogeneous conditions. Furthermore, the containers for standards and mobile phase and for sample

preparation and storage were always almost filled to capacity and sealed.

Regression analysis of the calibration curve from 1 [$y = (0.0167 \pm 0.0002)x - (0.0037 \pm 0.0018)$], 3 [$y = (0.0109 \pm 0.0004)x + (0.0209 \pm 0.0074)$], and 2 [$y = (27.98 \pm 0.28)x + (0.76 \pm 0.21)$], gave $r = 0.9964$, 0.9968 , and 0.9998 , respectively, as correlation coefficients ($n = 8$). These correlation coefficients are suitable, specially by considering the low value of the wavelength employed in the detection of the different substances (191 nm). All the other statistical parameters are adequate. These calibration curves have been shown to be linear over a wide concentration range: 1–25 mg/mL for 1, 1–30 mg/mL for 3, and 0.01–3 mg/mL for 2, with all the curves almost passing through the origin.

The validation study (Table 1) mentioned above showed intra- and interassay precision values better than 6.5% for the three assayed drugs. On the other hand, accuracy values were acceptable (<10.5%) for all the substances. As Table 2 shows, absolute recoveries (mean $\pm$ SD) for 1, 2, and 3 were $92.6 \pm 3.9\%$, $96.6 \pm 3.5\%$, and $90.8 \pm 4.1\%$, respectively.

When an accurate quantification of 2 is required in formulations containing high amounts of this drug, interferences with the matrix peak eluting at 3.8 min can be avoided by dissolving the sample in the mobile phase. This step is not necessary in routine analysis, because of the high molar absorptivity of 2.

Our proposed technique was applied to quantify the composition of pharmaceutical formulations containing 1–3, and mannitol and sorbitol as excipients. Figures 1 and 2 show the obtained chromatograms, corresponding to 1 and one of our formulations, respectively.

In formulations containing low dose of drug, when only the quantification of 1 and 2 is required, samples can be prepared by evaporation and further dissolution in absolute ethanol.

Conclusions

An analytical method for the quantitative determination of 1–3 has been developed and validated. The proposed method

is simple in sample preparation and offers adequate sensitivity, accuracy, and reproducibility. This method has been successfully employed for the analysis of tablets containing these drugs. The chromatographic assay allows an important decrease in the analytical time, in comparison with the microbiological techniques proposed by the most important pharmacopoeias. So, this HPLC method can be used in routine analysis.

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